

Disclosure Day

The compact proof that OPH is a simulation theory of physical constants

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Headline equations.

$$\mathfrak{U}_{\text{OPH}} \in \text{Fix}(T)$$

$$P_\star = \Gamma(P_\star), \quad N_\star = \mathcal{C}(N_\star)$$

$$|g'_\star| = \left(\frac{N_\star}{\pi}\right)^{-1/48}, \quad \frac{v}{E_{\text{cell}}} = |g'_\star|^{4P_\star} = \left(\frac{N_\star}{\pi}\right)^{-P_\star/12}$$

$$\frac{G_{\text{geom}}}{\ell_\star^2} = \frac{P_\star}{4(P_\star/4)} = 1$$

$$G_{\text{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\text{Cs}}^2} \varepsilon_{\text{Cs}}^2$$

The table below gives the output surface. The proof then shows the short route from OPH closure to the constants and to gravity: two fixed points, a 24-tick bridge, one cancellation, and one clock conversion.

1. The outputs first

Start with the receipts. A theory that claims to compute nature should leave numbers on the table before it asks for patience. OPH computes these rows from the same closure architecture; the gravity row below is a four-line cancellation once the pixel branch is fixed. In the table, m denotes a particle mass, M a boson pole mass, θ_{ij} neutrino mixing angles, Ω density fractions, H_{dS} the de Sitter static-patch rate, $B_\star$ an inverse-area display, N_c color count, N_g generation count, τ_p proton lifetime, and λ_c the dark-sector finite-thickness coefficient. Each row uses the branch normalization defined in the cited OPH papers.

Lane	OPH-derived value	Comparison or interpretation	Residual	Details
Simulation substrate	$\mathfrak{U}_{\text{OPH}} \in \text{Fix}(T)$	The universe is the stable fixed point of observer-overlap repair.	structural	O1,O2
Local pixel	$P_\star = 1.6309682094039593 \dots$	Unique local cell-area and edge-entropy fixed point.	derived	O3,O4,O5
Local cell entropy	$\bar{\ell}_{\text{shared}} = P_\star/4 = 0.407742052350989825 \dots$	Shared edge entropy assigned to one observer cell by the same fixed point.	derived	O3,O4
Global capacity	$N_\star \simeq 3.31 \times 10^{122}$	Closed-screen record capacity on the selected branch.	derived	O1,O4
24-tick repair factor	$ g'_\star = (N_\star/\pi)^{-1/48}$	One tick of the solved 24-step global repair rhythm.	derived	O4,O5
Hierarchy resonance	$v/E_{\text{cell}} = (N_\star/\pi)^{-P_\star/12}$	Weak scale as the local projection of the global repair cycle.	derived	O4,O5
Higgs naturality	$\epsilon_H = 0, \quad \epsilon_H \in [0, 0]$	OPH naturally solves the hierarchy row: the selected source-to-Higgs normal form has zero RG/coarse-graining defect.	exact	O4,O5
Clock bridge	$\varepsilon_{\text{Cs}} = 3.11393051342 \dots \times 10^{-33}$	The OPH length quantum expressed in the cesium-clock unit chart.	derived	O4
	$\gamma_\star = 4.95597433655 \dots \times 10^{-34}$			
	$B_\star = 3.60787391468 \dots \times 10^{70} \text{m}^{-2}$ $\ell_\star^2 = 2.61228030237 \dots \times 10^{-70} \text{m}^2$			
Area quantum and cell area	$a_{\text{cell}} = 4.26054612722 \dots \times 10^{-70} \text{m}^2$	The OPH area unit and physical area of one screen pixel.	derived	O3,O4
Newton coupling	$G_{\text{geom}} = \ell_\star^2$	One OPH pixel cancellation gives the geometric coupling; CODATA 2022 lists $G = 6.67430(15) \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$.	$\approx 0\sigma$	S1,O3,O4
	$G_{\text{SI}} = 6.674299995910528 \times 10^{-11}$			
Cosmological constant	$\Lambda_{\text{CRC}} \approx 1.09 \times 10^{-52} \text{m}^{-2}$	Planck 2018 late-universe Λ CDM comparison at displayed precision.	hit	S5,O1,O4
Static-patch rate	$H_{\text{dS}} \approx 55.759940256 \text{ km s}^{-1} \text{Mpc}^{-1}$	De Sitter static-patch display, distinct from the fitted local H_0 .	derived	O1,O4
Hubble benchmark	$H_0 = 67.4 \text{ km s}^{-1} \text{Mpc}^{-1}$	Planck 2018 reports $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{Mpc}^{-1}$.	0σ	S5,O1,O4
Lorentz group	$\text{Conf}^+(S^2) \cong \text{SO}^+(3,1)$	The support-visible screen branch reconstructs Lorentz kinematics.	structural	O4
Causal speed	$c = 299792458 \text{ m s}^{-1}$	Exact SI definition used for the Lorentz speed display.	exact	S3,O4
Einstein branch	Jacobson-type local Einstein equation	Local horizon thermodynamics and observer overlap select the Einstein equation branch.	structural	O3,O4
Gauge quotient	$\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y / \mathbb{Z}_6$	The Standard Model gauge quotient drops out of the selected compact gauge branch.	exact	S6,O4,O5
Hypercharge, color, generations	one-generation hypercharge lattice $N_c = 3, \quad N_g = 3$	Charge lattice, color count, and family count are branch readouts.	exact	S6,O4,O5
Protected carriers	$m_\gamma = 0$ $m_g = 0$	Photon, gluon, and graviton masslessness are protected by the selected gauge and diffeomorphism branches.	structural	S6,O4,O5
Gauge proton decay	$m_{\text{grav}} = 0$ $\tau_p^{(\text{gauge})} = \infty$	The selected gauge quotient has no simple-GUT gauge-boson proton-decay channel.	allowed by limits	S6,O5
Fine-structure source trunk	$\alpha_{\text{source}}^{-1} = 136.994835164621649 \dots$	Source-side electromagnetic trunk before endpoint transport.	derived	O5,O6
Fine-structure endpoint	$\alpha^{-1}(0) = 137.035999177(21)$ $\alpha(0) = 0.00729735256433 \dots$	CODATA gives $\alpha^{-1} = 137.035999177(21)$.	0σ	S2,O5,O6
Electroweak source	$\alpha_2(m_Z) = 0.03377843630219015$ $\alpha_Y(m_Z) = 0.010131601067241624$	Weak and hypercharge couplings on the selected electroweak branch.	fit-profile	S6,O5
Electroweak scale	$v = 246.76711732749683 \text{ GeV}$ $M_W = 80.377 \text{ GeV}$	Derived weak scale in the selected convention.	derived	S6,O5
W/Z validation pair	$M_Z = 91.18797809193725 \text{ GeV}$	PDG 2024 gives $M_W = 80.3692 \pm 0.0133 \text{ GeV}$ and $M_Z = 91.1876 \pm 0.0021 \text{ GeV}$.	0.59σ 0.18σ	S6,O5

Lane	OPH-derived value	Comparison or interpretation	Residual	Details
Higgs/top branch	$m_H = 125.1995304097179$ GeV $m_t = 172.35235532883115$ GeV $m_e = 0.00051099895$	PDG 2024 gives $m_H = 125.20 \pm 0.11$ GeV and direct $m_t = 172.57 \pm 0.29$ GeV.	-0.004σ -0.75σ	S6,S7,O5
Charged leptons	$m_\mu = 0.1056583755$ GeV $m_\tau = 1.7769324651340912$ $u = 0.00216, d = 0.00470$	PDG lepton summary gives the displayed central values.	$\leq 0.03\sigma$	S8,O5
Quark sextet	$s = 0.0935, c = 1.273$ $b = 4.183, t = 172.35235532883115$ 0.017454720257976796	GeV PDG quark summary gives the displayed light/heavy quark central values.	0σ shown	S7,O5
Neutrino masses	0.019481987935919015 eV 0.05307522145074924	Absolute individual masses are OPH branch outputs.	target	S10,O5
Neutrino sum	$\sum m_\nu = 0.09001192964464505$ eV	Planck 2018 plus BAO gives $\sum m_\nu < 0.12$ eV at 95% CL.	below limit	S5,S11,O5
Neutrino mixing display	$\theta_{12} = 34.2259^\circ, \theta_{23} = 49.7228^\circ$ $\theta_{13} = 8.68636^\circ, \delta = 305.581^\circ$ $\alpha_{21} = 153.618518^\circ$	NuFIT 6.0 gives global-fit profiles; the row is profile-comparison data.	profile	S9,O5
Majorana phases	$\alpha_{31} = 257.003241^\circ$ $\Omega_{\Lambda, \text{OPH}} = 0.684423332$	Absolute-phase neutrino observables are OPH branch outputs.	target	S11,O5
Compressed matter budget	$\Omega_A = 0.264114401$ $\Omega_m = 0.315484968$	Planck 2018 gives $\Omega_m = 0.315 \pm 0.007$ and $\Omega_\Lambda \simeq 0.685 \pm 0.007$.	$\Omega_m : 0.07\sigma$ $\Omega_\Lambda : -0.08\sigma$	S5,O1,O4,O7
Dark-sector coefficient	$\lambda_c = \exp(-P_*/24) = 0.934300639489386\dots$	Finite-thickness dark-sector susceptibility normalization.	derived	O7,O8

Comparison source keys.

S1: NIST CODATA 2022 G .

S2: NIST CODATA 2022 α^{-1} .

S3: NIST CODATA c , exact.

S4: NIST CODATA $\hbar$, exact.

S5: Planck 2018 cosmological parameters.

S6: PDG 2024 gauge and Higgs summary.

S7: PDG 2024 quark summary.

S8: PDG 2024 lepton summary.

S9: NuFIT 6.0 oscillation fit.

S10: KATRIN 2025 direct neutrino-mass limit.

S11: PDG 2024 neutrino-mass review.

Proof contract

The compact proof prints the gravity chain in full. It imports five certified branch records from the cited papers: the local pixel fixed point $P_\star$ from O3–O5, the global capacity fixed point $N_\star$ from O1/O4, the 24-tick hierarchy bridge from O4/O5, the cesium clock bridge from O4, and the particle/cosmology ledger rows from O1/O4–O8. Each imported record has a branch map, a selector, and a certificate in its source paper. Inside this PDF the dependency path is therefore:

$$\text{Fix}(T) \longrightarrow (P_\star, N_\star) \longrightarrow |g'_\star|, v/E_{\text{cell}} \longrightarrow G_{\text{geom}} = \ell_\star^2 \longrightarrow G_{\text{SI}}.$$

The local calculation has no adjustable gravitational parameter.

2. Yes, the universe is a simulation

OPH uses “simulation” in the hard mathematical sense. A physical world is an admissible record-update structure whose observer patches can read one another, repair disagreement, and return to the same structure after readback. The symbol $\mathfrak{U}_{\text{OPH}}$ denotes the selected OPH universe. The map T is the observer-overlap repair map. The notation $\text{Fix}(T)$ means the states returned unchanged by that map:

$$T(\mathfrak{U}_{\text{OPH}}) = \mathfrak{U}_{\text{OPH}}.$$

Read the equation as the simulation claim stripped to its mechanism. The universe is the state that survives its own observer-overlap audit. Finite records enter, overlap repair runs, contradictory branches die, and the returned object is the same object.

Operational reading

A software simulation advances a state by an update rule. OPH replaces the external programmer with a closed consistency rule: only observer records that can repair into a shared fixed point survive as physics. That is why the theory can speak about a simulated universe without adding a machine outside the universe.

The long papers prove the reconstruction theorems. *Reality as Consensus Protocol* gives the consensus mechanism; *Observers Are All You Need* gives the observer and record architecture; *Recovering Relativity and Standard Model Structure from Observer Overlap Consistency* gives the compact spacetime, gravity, and gauge branch; *Deriving the Particle Zoo from Observer Consistency* gives the quantitative particle branch. The certificates stay in those papers. This note names the proof spine.

3. Two constants from closure

The compact OPH branch has two dimensionless constants:

$$P_\star \quad \text{and} \quad N_\star.$$

The symbol $P_\star$ is the local pixel fixed point. It says how much area and shared edge entropy one elementary observer cell carries. The symbol $N_\star$ is the global screen-capacity fixed point. It says how many record units fit on the closed observer-facing horizon. They are branch outputs.

The local constant is computed from the readback equation

$$P_\star = \Gamma(P_\star), \quad \Gamma(P) = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

Here P is a candidate pixel value, Γ is the local OPH readback rule, $\varphi = (1 + \sqrt{5})/2$ is the golden-ratio screen balance point, and $A_T(P) = \alpha_{\text{em}}^{-1}(0; P)$ is the inverse low-energy electromagnetic width emitted by the same candidate cell. The equation asks for a pixel whose geometric reading and electromagnetic reading agree when the cell is read from inside the branch. The two terms are the mechanism: φ is the self-similar screen balance, while $\sqrt{\pi}/A_T(P)$ is the field-width correction carried by that same cell. Closure means the sum reads back as P .

The selected value is

$$P_\star = 1.630968209403959324879279847782648941 \dots$$

The number is the unique fixed point in the physical pixel interval. A different decimal in the same branch basin changes the starting guess, not the endpoint.

The global constant is computed from the screen-capacity readback equation

$$N_\star = \mathcal{C}(N_\star).$$

Here N is a candidate horizon record capacity and $\mathcal{C}$ is the OPH closed-screen readback rule. The equation asks for a universe whose observer-facing horizon contains exactly the capacity that the universe itself reads back. A closed screen has no outside ruler. Its admissible capacity is computed by the records it supports.

The equivalent count-density display is

$$N_\star = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N] \simeq 3.31 \times 10^{122}.$$

The set Ω_N^{sc} contains the closed-screen normal forms at capacity N . The term $\log |\Omega_N^{\text{sc}}|$ counts how many such normal forms are available, while the subtraction of N charges the screen for its record budget. The operator MAR chooses the minimal admissible arithmetic representative of the maximizing branch. The count-density problem records the same selection: record availability and record cost close on the same value.

The fixed-point test

For both $P_\star$ and $N_\star$, the detailed papers prove a contraction certificate on the relevant branch interval I :

$$F(I) \subseteq I, \quad d(F(x), F(y)) \leq q d(x, y), \quad 0 \leq q < 1.$$

Here F is the relevant readback map, d is the distance between two candidate values, and q is the shrink factor. The inequalities are the certificate: readback stays inside the branch and pulls candidates closer together. Banach's fixed-point theorem gives one stable endpoint for the whole interval.

4. The 24-tick bridge relates the two constants

The two constants belong to one branch. The local pixel and the global screen meet in the hierarchy branch, where particle physics samples a fixed piece of the same repair rhythm that closes the horizon record.

The global screen radius ratio supplies the repair tick:

$$|g'_\star| = \left(\frac{N_\star}{\pi} \right)^{-1/48}.$$

Here $g'_\star$ is the one-tick contraction factor of the global repair cycle. The ratio $N_\star/\pi$ is the dimensionless squared horizon-radius capacity in OPH normalization. The exponent $1/48$ is $1/(2 \cdot 24)$: the factor 2 converts capacity to radius, and the integer 24 is the repair-cycle contraction count on this branch. The equation turns the enormous screen capacity into the shrink factor for one repair tick. A full horizon-to-cell transport is a radius contraction; the selected branch decomposes it into 24 homogeneous ticks.

The electroweak branch samples that tick through the local pixel:

$$\frac{v}{E_{\text{cell}}} = |g'_\star|^{4P_\star} = \left(\frac{N_\star}{\pi} \right)^{-P_\star/12}.$$

The symbol v is the Higgs vacuum scale. The symbol E_{cell} is the energy associated with one OPH cell area. The exponent $4P_\star$ is the electroweak projection of the local pixel onto the 24-tick repair rhythm. The weak scale is the local/global resonance value, with no bare Higgs mass chosen by hand. Substituting the one-tick factor into the $4P_\star$ electroweak projection gives $-4P_\star/48 = -P_\star/12$, so the same formula is visible from the local pixel side and the global capacity side.

The bridge is often printed as

$$\Pi_{\text{EW}}(P, N) = \frac{24\pi}{\alpha_U(P) \log(N/\pi)}, \quad \Pi_{\text{EW}}(P_\star, N_\star) = 4P_\star.$$

Here Π_{EW} is the electroweak tick-projection ratio and $\alpha_U(P)$ is the unified-coupling value emitted by the pixel branch. The equation is the matching condition between the source-side transmutation exponent and the global repair tick. The numerator 24π is the 24-tick cycle with angular normalization; the denominator measures how the unified coupling and global capacity logarithm set the available transmutation phase.

5. Gravity without units

The gravity proof is short because $P_\star$ cancels. OPH first derives the Newton coupling as a geometric conversion between cell area and shared-edge entropy:

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

Here G_{geom} is Newton's coupling in OPH geometric units, a_{cell} is the area of one observer cell, and $\bar{\ell}_{\text{shared}}$ is the shared edge-entropy weight of that cell. The equation makes gravity the conversion factor between an area count and the edge entropy an observer patch shares with its neighbors. The factor 4 is the quarter-area normalization from horizon entropy and the OPH edge-entropy branch.

The selected pixel gives

$$a_{\text{cell}} = P_\star \ell_\star^2, \quad \bar{\ell}_{\text{shared}} = \frac{P_\star}{4}.$$

The symbol $\ell_\star$ is the OPH length quantum, so $\ell_\star^2$ is the OPH area quantum. The first equation assigns physical area to one cell; the second assigns the same cell its shared entropy weight. Both use $P_\star$, because $P_\star$ is exactly the dimensionless pixel ratio.

Substitution gives the dimensionless gravity result:

$$\frac{G_{\text{geom}}}{\ell_\star^2} = \frac{a_{\text{cell}}/\ell_\star^2}{4\bar{\ell}_{\text{shared}}} = \frac{P_\star}{4(P_\star/4)} = 1.$$

The pixel branch fixes every symbol in this line. Divide the geometric Newton coupling by the OPH area quantum and the ratio is one. The same $P_\star$ measures cell area and shared edge entropy, so the pixel value cancels instead of becoming an adjustable gravitational constant.

Thus

$$G_{\text{geom}} = \ell_\star^2.$$

The local gravity theorem is one line. The numerical row is unit conversion.

6. Gravity as a number in SI units

SI units enter through a clock bridge whose inputs are the unit chart and the cesium branch. OPH writes the bridge as

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}.$$

The symbol $\gamma_\star$ is the dimensionless ratio between the OPH length $\ell_\star$ and the distance light travels during one cesium-clock tick. The symbol $\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ s}^{-1}$ is the exact SI cesium frequency, c is the exact speed of light, and ε_{Cs} is the OPH cesium clock gap emitted by the atomic clock branch. This expresses the OPH length quantum in human clock units. The 2π appears because the clock branch uses angular phase while SI frequency counts cycles.

Solving for the length quantum gives

$$\ell_\star = \frac{c \varepsilon_{\text{Cs}}}{2\pi \nu_{\text{Cs}}}.$$

This turns a dimensionless clock gap into a length. Light travels c/ν_{Cs} meters per cesium cycle, and the OPH phase gap supplies the fraction of that cycle.

The geometric-to-SI conversion is

$$G_{\text{SI}} = \frac{c^3}{\hbar} G_{\text{geom}}.$$

Here G_{SI} is Newton's constant in $\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, and $\hbar$ is the reduced Planck constant. The equation converts the OPH area coupling into the SI Newton coupling. The usual Planck-area relation is $\ell_P^2 = \hbar G/c^3$, so a geometric area coupling becomes an SI gravitational coupling after multiplication by $c^3/\hbar$.

Combining the last two equations with $G_{\text{geom}} = \ell_\star^2$ gives

$$G_{\text{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\text{Cs}}^2} \varepsilon_{\text{Cs}}^2.$$

This computes the SI Newton constant from the OPH clock gap and exact SI unit definitions. The powers are just book-keeping: $\ell_\star^2$ contributes $c^2 \varepsilon_{\text{Cs}}^2 / (4\pi^2 \nu_{\text{Cs}}^2)$, and the area-to-SI conversion contributes $c^3/\hbar$.

The selected clock branch emits

$$\varepsilon_{\text{Cs}} = 3.1139305134160128239 \dots \times 10^{-33}, \quad \gamma_\star = 4.9559743365484194637 \dots \times 10^{-34}.$$

ε_{Cs} is the branch output. γ_* is the same output in the length chart.

Substitution gives

$$G_{\text{OPH}} = 6.674299995910528 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

The comparison with laboratory G happens after the derivation. The fixed-point equations above take their inputs from closure records alone.

Paper map

O1: Observers Are All You Need.

O2: Reality as Consensus Protocol.

O3: Screen Microphysics and Observer Synchronization.

O4: Recovering Relativity and Standard Model Structure from Observer Overlap Consistency.

O5: Deriving the Particle Zoo from Observer Consistency.

O6: Fine Structure Constant Derivation.

O7: OPH Dark Matter Paper.

O8: Chi-Nu Susceptibility Bounds.

How to read this PDF

The ledger is the claim surface; the proof sections give the compact route through it. The mathematical load is carried by the cited core and extra papers, where each branch map, interval, selector, contraction, normal form, and numerical certificate is specified. The compact claim is: simulation first, two computed dimensionless constants, 24-tick local/global resonance, $G_{\text{geom}} = \ell_*^2$, and the SI gravity number from the cesium clock bridge.